Derivability equals inconsistency with the negated sentence

The Open Logic Project contributors

openlogic:first-order-logic/tableaux:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-tableaux/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof argues by transformation of closed tableaux in prose; no tableau environment occurs in the counted span

Editorial note. The label prop:prov-incons names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Allowed standard prerequisites (editorial).

- Signed tableau rules for negation. From the signed formula True (not A) the True-not rule derives False A; from False (not A) the False-not rule derives True A.. The forward direction converts the witness tableau with the True-not rule; the reverse direction closes branches with the False-not rule.
- Inconsistency via closed tableaux. A finite signed assumption set is inconsistent iff it has a closed tableau.. Both directions of the equivalence are phrased as existence of closed tableaux for the corresponding signed assumption sets.

Proposition 0.1. $\Gamma \vdash A$ *iff* $\Gamma \cup \{\neg A\}$ *is inconsistent.*

Proof. First suppose $\Gamma \vdash A$, i.e., there is a closed tableau for

$$\{\bot(A), \top(B_1), \ldots, \top(B_n)\}$$

Using the $\top\neg$ rule, this can be turned into a closed tableau for

$$\{\top(\neg A), \top(B_1), \ldots, \top(B_n)\}.$$

On the other hand, if there is a closed tableau for the latter, we can turn it into a closed tableau of the former by removing every formula that results from $\top\neg$ applied to the first assumption $\top(\neg A)$ as well as that assumption, and adding the assumption $\bot(A)$. For if a branch was closed before because it contained the conclusion of $\top\neg$ applied to $\top(\neg A)$, i.e., $\bot(A)$, the corresponding branch in the new tableau is also closed. If a branch in the old tableau was closed because it contained the assumption $\top(\neg A)$ as well as $\bot(\neg A)$ we can turn it into a closed branch by applying $\bot\neg$ to $\bot(\neg A)$ to obtain $\top(A)$. This closes the branch since we added $\bot(A)$ as an assumption. □